



Spend Intentionally:

cutting LLM costs without cutting
quality with Kong AI Gateway



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AGENDA

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cost-optimization with Kong

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The enterprise AI cost challenge

Rapid GenAI adoption is pushing enterprise budgets to their limits, demanding smarter traffic governance and policy control

ACADEMIC INSIGHT

"Intelligent LLM routers learn to predict query difficulty. By dynamically routing simple queries to smaller models and reserving complex ones for frontier models, we can maintain 95% of GPT-4's performance while cutting public API costs by up to 85%."

- [UC Berkeley & Anyscale Research](#)

ANALYST PERSPECTIVE

"Enterprise AI budgets are coming under greater scrutiny, with increased focus on usage efficiency, cost control, and measurable outcomes. As usage-based pricing becomes harder to predict, buyers will turn to platforms that help them choose the right tools, monitor performance, enforce policy, and keep costs under control."

- [Arunasree Cheparthi, Gartner](#)

ENTERPRISE REALITY

"93 percent of organizations report exceeding their AI budgets... Businesses that develop capabilities to optimize spend, improve accountability, and redirect savings could reduce their AI costs by 20 to 30 percent."

- [Pankaj Sachdeva, Wasim Lala, McKinsey & Company](#)

The background of the slide features a vibrant green, abstract pattern resembling the fronds of a palm or a similar tropical plant. The pattern is composed of numerous fine, radiating lines that create a sense of depth and movement. The colors range from a bright, almost neon green to a darker, forest green, with some areas appearing to have a slight glow or gradient. The overall effect is organic and modern.

The fix isn't rewriting application code - it's mediating every request at the AI Gateway



the solution

Mediation at the AI Gateway

[Shadow AI]

[Client App] -----(Hard-coded Credentials & Targets)-----> [Frontier LLMs] (\$\$\$\$)

[Kong AI Gateway Mediation]

[Client App] --(Unified Route)--> [Kong AI Gateway] --(Optimized)--> [Frontier LLMs] (\$)
|--> [Redis Cache]
|--> [Compressor Service]
|--> [FinOps Firewall]

Unified Endpoint: Decouple application code from backend LLM providers.

Dynamic Credentials: Stop distributing API keys to local developer environments.

Operational Control: Inject caching, compression, routing, and rate-limiting at the proxy tier.



Live demo



Demo 1

Semantic Caching

Client	Kong AI Gateway	Embeddings API	Redis VectorDB
---- 1. POST "Cook filet" ->			
	---- 2. Vectorize prompt ->		
	<--- 3. Return Vector -----		
	---- 4. Cosine similarity query ----->		
	<--- 5. Match Found: "Steak Recipe" (Score: 0.95) -		
<-- 6. Cached Answer (5ms)--			

Plugin: ai-semantic-cache

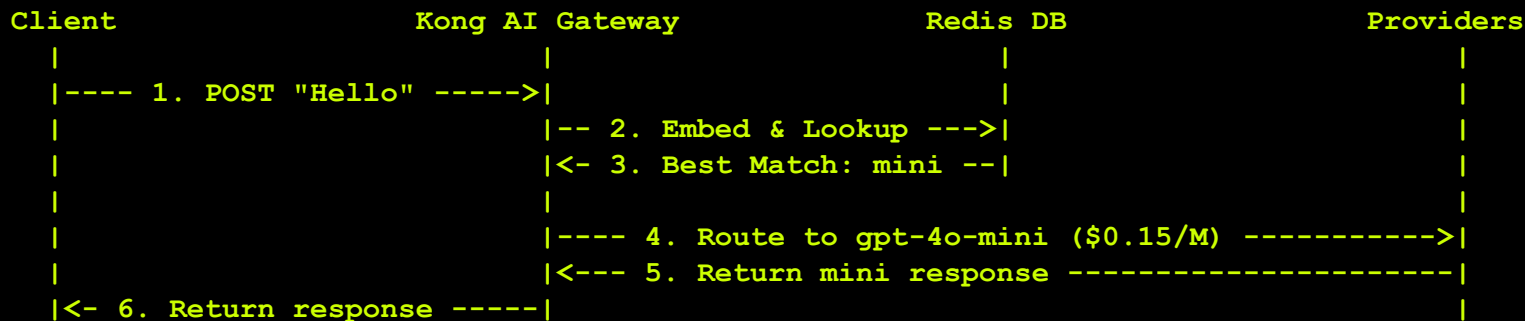
Concept: Match prompts by meaning rather than exact strings.

Outcome: Returns cached answers in <10ms at \$0 provider cost—bypassing the upstream LLM entirely.



Demo 2

Semantic Routing



Plugin: ai-proxy-advanced (algorithm: semantic)

Concept: Route requests to different models based on the prompt's complexity.

Outcome: Greetings are routed to gpt-4o-mini (\$0.15/M tokens), while advanced engineering logic is promoted to gpt-4o (\$2.50/M tokens).

Demo 3

Intelligent Model Routing

Client	Kong (Datakit)	Kong (Classifier Route)	OpenAI (gpt-4o-mini)
--- 1. POST Chat --->			
	--- 2. Call Classifier ->		
		--- 3. Prepend Router prompt->	
		<-- 4. Return "gpt-4o-mini"---	
	<-- 5. Return target ----		
	--- 6. Mutate payload (model: gpt-4o-mini) ----->		
	<-- 7. Execute Chat completion -----		
<- 8. Chat Answer ---			

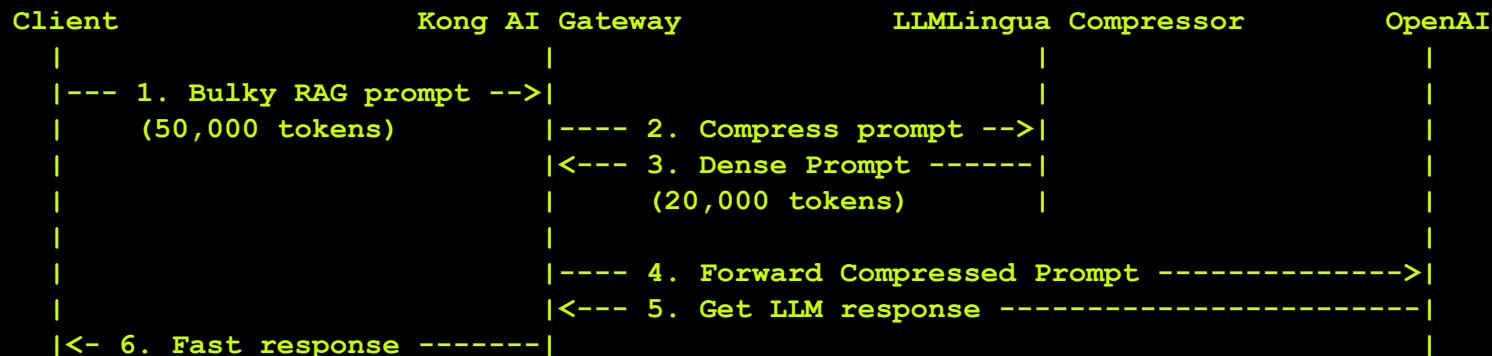
Plugins: datakit + ai-prompt-decorator

Concept: Use a fast "classifier model" to explicitly choose and rewrite the final target model in flight.

Outcome: Complete gateway-tier serverless orchestration with zero code.

Demo 4

Prompt Compression



Plugin: ai-prompt-compressor

Concept: Prune low-information filler words from massive prompts in flight.

Outcome: Cuts token consumption and billing by up to 60% while preserving close to 100% reasoning accuracy.

Demo 5

AI (cost) rate Limiting

Client	Kong AI Gateway	Redis Store	OpenAI
---- 1. POST chat ----->			
	--- 2. Fetch current cost>		
	<-- 3. Return remaining --		
	-- [Budget OK?]		
	Yes -> Call OpenAI ----->		
	No -> Return HTTP 429 (Blocked!)		
<- 4. Response / 429 -----			

Plugin: ai-rate-limiting-advanced (tokens_count_strategy: cost)

Concept: Block requests when users exceed real-time dollar-spend ceilings instead of connection counts.

Outcome: Bulletproof budget protection against infinite inference loops.

Demo 6

Model quality with LLM-as-a-Judge

```
Client          Kong AI Gateway          gpt-4o-mini (Primary)          gpt-4o-mini (Judge)
|               |               |               |
|-- 1. POST chat ---->|               |               |
|               |--- 2. Call Primary ----->|               |
|               |<-- 3. Fast Answer -----|               |
|<- 4. Return Answer -|               |               |
|               |               |               |
|               |--- 5. (ASYNC) Call Judge with original prompt & mini reply->|               |
|               |<-- 6. Return Score (1-100) -----|               |
|               |               |               |
|               |--- 7. Publish `llm_accuracy: 95` to logs/Collector ----->|               |
```

Plugin: ai-llm-as-judge

Concept: Asynchronously grade cheap model responses in production using a cost-efficient judge model.

Outcome: Data-driven proof to confidently justify migrating applications to cheaper models.



Demo 7

Visibility to your spend over Kong Analytics

```
Kong Data Plane (Inference Logs) ---> OTel Collector ---> Grafana / Loki Dashboard
      \
        ---> Kong Konnect (Central AI/LLM Dashboards)
              |--> Spend by Consumer / API Key
              |--> Token Latency by Provider
              |--> Token Volume by Model
```

Feature: Konnect Analytics - AI/LLM Summary

Concept: Real-time visibility into overall token throughput, latencies, and dollar spend.

Outcome: Group, pivot, and generate custom reports by Service, Route, Consumer, and Model.

References



References

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- [The cost of intelligence: How CIOs can manage AI demand at scale \(McKinsey & Company\)](#)
- [RouteLLM: Learning to route LLMs with preference data \(research paper\)](#)
- [Kong AI semantic cache plugin](#)
- [Kong AI proxy advanced plugin](#)
- [Kong semantic routing examples](#)
- [Kong AI prompt compressor plugin](#)
- [Kong AI rate limiting advanced plugin](#)
- [Kong AI LLM as a Judge plugin](#)
- [Kong Datakit plugin](#)
- [Kong AI prompt decorator plugin](#)
- [Kong post-function plugin](#)
- [Kong model-based routing cookbook](#)
- [Kong how-to guide: compare llm models accuracy](#)
- [deck CLI installation & gateway GitOps hub](#)





Questions?





Thank you!



Thank you!

Ready for what's next?
Let's talk

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Kong



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